Piece Yen

Area of Interest

Auditory Neuroscience Neural Plasticity pieceyen@gmail.com

RESEARCH EXPERIENCES

School of Biosciences, University of Sheffield
Research Associate, Apr 2023 – Sep 2025 (Dr. Stuart Johnson)

- Investigating the physiology of accelerated progression of age-related hearing loss after temporary noise-induced hearing loss in early-adult.

Institute of System Neuroscience, National Tsing Hua University
Research Associate, Oct 2013 - Sep 2015 (Dr. Chung-chuan Lo)

Title: Multi-scale interactions between pre- and post-synapses shapes transcranial magnetic stimulation-induced long-term plasticity (grant-funded).

- Developed a calcium-dependent plasticity (CaDP) model at single synapse and network scale.
- Predicted the plasticity induced by novel TBS pattern.

Title: Short-term spatial memory in un-trained Drosophila Melanogaster.

- Designed and built a customized experimental apparatus.

EDUCATION

2018 – 2022 The University of Sheffield, Sheffield, United Kingdom
Doctor of Philosophy in Biomedical Sciences

Title: The influence of non-sensory cells on the development of mammalian inner hair cells.

- My study provided evidence that ATP signaling, without potassium secretion, from the non-sensory cells is enough to drive the functional maturation in cochlear hair cells.

2010 – 2012 National Tsing Hua University (NTHU), Taiwan
Master of Science in Systems Neuroscience, 4.1/4.3

Title: A Spiking Neural Network Model of Motor Cortex for Transcranial Magnetic Stimulation-Induced Synaptic Plasticity.

- Built a large-scale neural network model of primary motor cortex to simulate synaptic plasticity.
- Developed and studied the candidate mechanism of TMS-induced synaptic plasticity.

2006 – 2010 National Tsing Hua University (NTHU), Taiwan
Bachelor of Science in Life Science, 76.89/100

SKILLS

Critical thinking In-vivo hearing tests in mouse model Patch-clamp electrophysiology

micro-dissection of ex-vivo inner ear tissue Inducible Cre/LoxP workflow in transgenic mouse strains
Immunofluorescent labelling Super-resolution shadow imaging of inner ear tissue
Airyscan super-resolution confocal system Programming language (Python) Scientific writing
Large-scale neural network modeling

ACADEMIC AWARDS

2024 – Wellcome Trust Biomedical Vacation Scholarships (as project host)
2012 – Best Paper Award, SEMBA Conference, Taichung, Taiwan
2010 – Master Studentship (100,000 NTD) in College of Life Science, NTHU

PUBLICATION

Yen P, Carlton AJ, O'Connor A, Johnson SL, Jeng J-Y. Aging Alters Hair Cell Physiological Properties in Mice with Late-Onset Age-Related Hearing Loss 2025:2025.10.24.684339. https://doi.org/10.1101/2025.10.24.684339.

Yen P*, Manca M*, Spaiardi P, Russo G, Giunta R, Johnson SL, Marcotti W, Masetto S. Current Response in Ca V 1.3-/- Mouse Vestibular and Cochlear Hair Cells. Front Neurosci. 2021 Dec 8;15:749483. doi: 10.3389/fnins.2021.749483. PMID: 34955713; PMCID: PMC8694397. (**joint first authorship*)

Jeng JY, Ceriani F, Hendry A, Johnson SL, **Yen P**, Simmons DD, Kros CJ, Marcotti W. Hair cell maturation is differentially regulated along the tonotopic axis of the mammalian cochlea. J Physiol. 2020 Jan;598(1):151-170. doi: 10.1113/JP279012. Epub 2019 Dec 21. PMID: 31661723; PMCID: PMC6972525.

Yu-yang Yen; Lee-ray Chen; Ying-zu Huang; Chung-Chuan Lo, "Models of cortico-basal ganglia circuits and synaptic plasticity for transcranial magnetic stimulation," *Complex Medical Engineering (CME), 2012 ICME International Conference on*, vol., no., pp.96,100, 1-4 July 2012 doi: 10.1109/ICCME.2012.6275678

REFEREES

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